

CRE Development Potential and the Selection of Opportunity Zones*

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February 9, 2026

Abstract

Place-based policies are often caught between two potentially conflicting aims: (i) directing aid to needy communities and (ii) spurring investment. We study this tradeoff in the context of the Opportunity Zones (OZ) program. Leveraging unique phase-level microdata on commercial construction projects, we show that US state governors prioritized designating tracts where construction projects were already being planned. About two-thirds of the greater construction growth in OZs can be attributed to this selection. States prioritizing tracts with greater investment opportunities observed larger construction increases in designated tracts. We calibrate a structural model to quantify the effects of the program and examine counterfactuals under alternative preferences or eligibility criteria.

Keywords: opportunity zones, commercial real estate, construction, time-to-plan

JEL Classification: R23, R32, R58

*The views expressed in this paper are solely those of the authors and do not necessarily reflect the opinions of the Federal Reserve Board or the Federal Reserve System.

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1. INTRODUCTION

Place-based policies often target distressed areas in the spirit of community redevelopment (Hanson et al., 2025). While such policies can be desirable on equity grounds (Gaubert et al., 2025), channeling aid to highly distressed regions may be hindered by a lack of viable investment opportunities in such neighborhoods (Glaeser and Gyourko, 2005). Policy makers thus face a trade-off between supporting areas that are highly distressed and those that are more likely to respond to program incentives. These considerations also complicate analysis of place-based policies since program effects need to be separated from selection effects (e.g., the designation of markets with severe distress or high economic potential).

The Opportunity Zones (OZ) program offers an illuminating case study of this tension. Enacted as part of the 2017 Tax Cuts and Jobs Act, the program allowed governors to designate up to 25 percent of eligible low-income or high-poverty census tracts for preferential capital gains tax treatment for qualifying investments.¹ Although the policy was intended to channel private capital toward distressed communities, state governors retained substantial discretion over which tracts to select from among those eligible. We study how tracts were selected in light of these tradeoffs, and the implications of this behavior for evaluating the program's effects.

The key challenge is that states likely select OZs partly based on difficult-to-observe factors related to construction potential. For example, Corinth and Feldman (2024) show that tax incentives are largest for projects that are profitable enough to occur without the program and Corinth et al. (2025) show that OZs attract more investment in high-growth regions. These factors might induce governors to designate tracts that would have had more activity even without the OZ program. However, without the ability to measure this construction potential, it is difficult to quantify the extent to which such selection matters.

In this paper we conduct three pieces of analysis to overcome this difficulty and quantify the effects of OZ designation in the presence of selection. First, we analyze how OZs

¹Eligibility was defined by a poverty rate exceeding 20 percent or median income below 80 percent of the area median (the higher of the state or local MSA median income). Additionally, up to 5 percent of designated zones could be contiguous tracts that did not meet these criteria but bordered qualifying areas and had a median income not exceeding 125 percent of their designated neighbor. We refer the reader to Corinth and Feldman (2024) for a thorough discussion of the tax treatment of OZs.

were selected and how construction activity responds to OZ designation using data that are uniquely suited to measuring development potential. Specifically, we leverage detailed project-level data from Dodge Data & Analytics (Dodge) that tracks commercial construction activity starting from early stages of the construction process. Critical to this study, the data includes projects in the planning phase—those where architects have been retained and designs are underway but before permits are filed—thus providing visibility into potential construction that is useful for distinguishing genuinely new investment stimulated by the OZ program from projects that were already underway.² We use this data to demonstrate that while census tracts designated as OZs had substantially more construction than other eligible tracts, the difference drops by nearly two-thirds (from 0.46 projects per tract to 0.18) when preexisting planning activity is controlled for.

Second, we exploit cross-state variation in how opportunity zones were selected to examine (i) how states balanced the trade-off between targeting areas with distress and those with development momentum and (ii) how differences in selection relate to program outcomes. We show that states that prioritized designating tracts with greater development potential—namely, those that disproportionately selected tracts with more pre-OZ planning activity or publicly indicated that development potential was an important consideration in designated tracts—observed larger construction gains in OZ-designated tracts. In fact, we only find a material increase in construction for OZ-designated tracts in states above the median in development potential prioritization.³

In the final part of the paper, we estimate a structural model of OZ selection that formalizes the tension between targeting poverty and maximizing development impact. Governors prefer to designate tracts that have more development potential and higher poverty, but differ in the relative preference for each. We jointly model OZ designation decisions and construction outcomes, accounting the heterogeneous effects of OZ designation by development potential, cross-state variation in governor preferences, and unobserved variation in development potential.

²Dodge’s data is the most comprehensive data that we are aware of on commercial construction projects, and is used by the Census in constructing their sampling frame for their commercial Construction Put in Place statistics. See [Glancy et al. \(2025\)](#) for more details on the data.

³This finding does not mean that the relationship between OZ designation and construction activity is entirely due to selection bias. It is possible that states that prioritized designating high-poverty areas selected tracts that would have had lower construction activity without the OZ program.

The estimation of the model aligns closely with the empirical results, and yields several new insights. First, it allows us to estimate the effect of the OZ program accounting for selection on unobserved development potential. We estimate that OZ designation increased construction activity by about 13 percent in designated tracts. While this effect is economically meaningful, it is somewhat below the reduced form estimate of 22 percent, reflecting the effects of selection on unobservable development potential. Second, the model allows us to quantify the tradeoff between targeting areas with high poverty vs. high development potential. An increased prioritization of distressed tracts that raises the average poverty rate for designated tracts by two percentage points reduces the effect of OZ designation by about 0.01 properties per designated tract, reflecting the shift in designation to tracts with fewer viable projects. Third, we use the model to examine the effect of changing eligibility criteria to the revised criteria used in the One Big Beautiful Bill Act (OBBBA). We show the effect of stricter tract income and poverty thresholds has a similar effect to increasing state prioritization of distressed tracts, namely OZs shift to higher poverty areas, but at the cost of slightly lower program effects.

Our work broadly relates to the empirical literature on the selection process for opportunity zones. [Gelfond and Looney \(2018\)](#) document early evidence that states systematically selected tracts with higher income and home price growth compared to other eligible areas. [Eldar and Garber \(2023\)](#) find that while governors selected tracts with higher distress levels, they simultaneously favored tracts on upward economic trajectories, that supported the governor electorally, and where potential investors had donated to gubernatorial campaigns. [Frank et al. \(2022\)](#) demonstrate that states favored tracts whose representative's political affiliation matched that of the state's governor, noting that proportional allocation or local nomination mitigated this bias. We add to this work by documenting selection based on pre-OZ construction planning activity, and examining differences across states in how investment potential and distress were prioritized.

A likely reason for states to prioritize tracts on upward trajectories is that the program would affect more projects and stimulate more additional investment (assuming high growth areas have more potential projects at the margin of occurring). Consistent with this idea, [Corinth et al. \(2025\)](#) shows that opportunity zones attracted more investment in areas with more pre-existing investment. [Kennedy and Wheeler \(2022\)](#) find that early OZ investment concentrated in merely 16 percent of designated zones, predominantly in

neighborhoods with pre-existing upward trends in population, income, and home values.⁴ [Coyne and Johnson \(2023\)](#), using Treasury tax data through 2020, confirm that investment flowed disproportionately to zones with higher median household incomes, educational attainment, and pre-existing growth trends, with \$48 billion in qualified opportunity fund assets concentrated in areas already receiving investment. Our work provides additional evidence of heterogeneous effects of OZ designation—investment appears to respond more to designation in tracts where states prioritize investment potential more—and constructs a model that allows us to quantify the tradeoff between aiding distressed areas and stimulating more construction.

Other estimates of the treatment effects of opportunity zones have been mixed, with the findings dependent on timing and methodology ([Corinth and Feldman, 2024](#)). [Arefeva et al. \(2024\)](#) find sizable employment effects of OZ designation, but [Freedman et al. \(2021\)](#); [Atkins et al. \(2023\)](#) little effect on employment or job postings. There is more evidence of OZ designations affecting real estate, which accounted for about 75% of OZ investment ([Corinth et al., 2025](#)).⁵ [Wheeler \(2022\)](#) documents a 20 percent increase in the probability of new development activity in designated zones and [Sage et al. \(2023\)](#) finds greater commercial property price appreciation in designated tracts, especially for vacant land. However, [Chen et al. \(2023\)](#); [Alm et al. \(2024\)](#) find no effect on residential and commercial real estate prices, respectively. Observing projects before they enter the permitting phase allows us to better distinguish development that was already underway from genuinely new activity stimulated by OZ designation.

2. DATA AND SUMMARY STATISTICS

We start by discussing the tract-level data, including OZ status, construction and planning activity, and controls for tract level analysis. We then discuss state-level variables measuring differences in how states selected OZs.

⁴[Fikri et al. \(2023\)](#) synthesize multiple studies and also find a high concentration of investment in designated zones.

⁵Real estate is particularly well-suited to OZ investment because of the clearer eligibility criteria and the fact that taxable returns are concentrated in capital gains, which receive favorable treatment by the program.

2.1. *Tract data*

Our main data are project-level information from Dodge Data and Analytics, which we supplement with tract-level information from the American Community Survey (ACS).⁶ The Dodge data provides comprehensive coverage of commercial construction projects across the United States, tracking projects from initial planning through completion (or abandonment if the project ends up being canceled). Critically for our analysis, Dodge captures projects at the planning stage—after an architect has been selected and design work has begun, but before building permits are filed. This early visibility into the development pipeline allows us to identify projects that were already being planned before OZ designation occurred, providing a unique measure of pre-existing development momentum that is unavailable in permit-based datasets. The data from Dodge also includes geographic information and information on the expected construction spending at the planning phase and then the construction spending at the start phase, as well as information on the property type and the building square footage.

We aggregate the Dodge project-level data, creating counts of planning starts, construction starts, and completions for both the pre-OZ period (before 2018) and the post-OZ period (2018-2022). Planning starts represent projects entering the design phase, construction starts indicate ground-breaking, and completions mark project finalization. We also track project abandonments during the planning phase—instances where planned projects fail to proceed to construction—which provide insight into marginal projects that might benefit from additional incentives.

Tract demographics, including median income and the poverty rate used to determine OZ eligibility, come from the 2015 5-year American Community Survey (ACS). Relative income is the eligibility criteria comparing the median tract income to the greater of the state or MSA median (i.e., the ratio of median income for a tract and that of the city or state). We also use ACS data on the share of occupied housing units that are renter occupied (Renter share) as a control since demand for multifamily housing might also contribute to designation decisions.

We obtain a list of designated and eligible tracts, including whether those eligible tracts are contiguous or independently eligible, from the IRS. Tracts are eligible to be an opportunity

⁶Millar et al. (2016) and Glancy et al. (2025) provide further details on the data.

zone if they (i) have a poverty rate of at least 20 percent (ii) have a relative income no greater than 0.8, or (iii) are contiguous with a designated tract meeting the first two criteria, and have a relative income under 1.25.

We also use information on 2018 employment counts by census tract from LODES workplace area characteristics, and a tract-level housing supply elasticity from [Baum-Snow and Han \(2024\)](#).⁷ To avoid restricting the sample, we fill in missing housing supply elasticity information (typically for tracts outside of MSAs) using the sample mean, and include an indicator for whether the variable is missing in our analysis.

Table 1 contains tract-level summary statistics. Column (1) is limited to tracts that were not OZ-eligible, column (2) is limited to designated tracts, and column (3) includes tracts that are eligible, but were not designated. In column (4), we report the difference in means between columns (2) and (3). While OZs had more construction activity (about 0.8 more plan starts per tract, and about 0.4 more construction starts and completions), they also had many characteristics likely to be associated with activity independent of designation: more employment, more planning activity before the program started, and a higher share rental units (relevant since multifamily construction was a major component of OZ investing).

The statistics in Table 1 point to potential motivations for designation. OZ tracts were more likely to have projects abandoned in the pre-OZ period, suggesting the presence of marginal projects that could benefit from extra inducements to go forward. Additionally OZs had higher poverty rates and lower relative incomes, suggesting a prioritization of areas in greater need of support.

2.2. State Heterogeneity in the Designation Decision

To understand heterogeneity in selection across states, we use three measures of the degree to which states strategically designated eligible tracts. Our first measure comes from [Frank et al. \(2022\)](#), and is an indicator for whether the state's selection process included some form of a proportional criteria (for example, proportionally across counties). States with proportional selection would have less discretion to allocate funds to areas with the greatest investment potential (for example, they could designate tracts within

⁷We map the [Baum-Snow and Han \(2024\)](#) measures to 2010 tract definitions using the cross walk from the Longitudinal Tract Database of [Logan et al. \(2014\)](#).

a county with greater potential, but not disproportionately designate tracts in faster growing counties). Table A.1 includes a list of states and an identifier for those that used proportional selection.

For our second measure, we impute the extent to which states targeted growing areas by estimating how much weight previous planning activity played in the selection decision. Specifically, we estimate a probit:

$$\Pr(\text{Designated}_i) = \Phi(\alpha_{s(i)} + \beta_{s(i)} \text{pre-OZ Plan Count}_i + \gamma X_i)$$

where i indexes tracts, and $\beta_{s(i)}$ is a state-specific sensitivity of selection to initial planning activity. This can be thought of as a random utility model, where $\alpha_{s(i)}$ reflects (negatively) a state’s reservation utility, and $\beta_{s(i)}$ reflects how much a governor values investment potential in designation. States that have a greater preference for investment potential in designation would be expected to have larger estimated responses to designation due to these selection effects being stronger. We refer to $\hat{\beta}_s$ as a state’s planning sensitivity.

For our third measure, we construct an LLM estimate of how much states prioritized investment potential in designating tracts. Specifically, we had GPT-5 (from OpenAI) analyze the factors different states considered in designating opportunity zones and score states from 1–5 on how much they considered investment opportunities in designating tracts.⁸ For example, Texas scored a 1 as “a multi-step process was used to identify eligible areas in particular need due to chronic unemployment, lower population density and significant economic disruptors, such as natural disasters.”⁹ Indiana had a score of 5 because OZs “were selected based on a combination of factors including existing economic development programs and local coordination, economic and community data, likelihood of attracting short- and long-term investment, and growing industry sectors within the community.”¹⁰ Figure A.1 shows that states that did not disproportionately

⁸We prompted GPT-5 with the prompt: “Write a report on the factors that different states considered in designating opportunity zones. For each state, provide a ranking from 1-5 characterizing how much states considered the presence of investment opportunities in designating tracts. Also provide a ranking from 1-5 on the confidence in your characterization. If there was some other factor that seemed critical to designation, please report that. Please provide supporting evidence for each ranking in the form of excerpts and links to where the information is coming from.” We constructed this measure as of August 13th, 2025, which provided us the output presented in Table 1 and the additional requested commentary.

⁹See: <https://gov.texas.gov/business/page/opportunity-zones>.

¹⁰See: <https://iedc.in.gov/program/indiana-opportunity-zones/overview>.

designate tracts with high initial planning almost exclusively had low scores (2 or below) by this investment prioritization metric.¹¹

3. EMPIRICAL RESULTS

Section 3.1 presents findings predicting tract-level OZ designation decisions and post-OZ construction activity. Section 3.2 examines how patterns differ by state selection criteria.

3.1. *Tract-level determinants of OZ designation and construction*

Table 2 presents our core empirical analysis of how pre-existing planning activity and other tract characteristics relate to opportunity zone designation and subsequent construction activity. The table reveals two key findings: first, that selection into the OZ program was systematically related to pre-existing development potential, and second, that controlling for this selection significantly reduces the estimated program effect. All specifications include state fixed effects, so estimates reflect differences either the probability of OZ selection or the number of construction starts in OZs relative to other OZ-eligible tracts in the same state. In this analysis, and throughout the remainder of our empirical analysis, standard errors are clustered at the state level.

Column (1) estimates a probit model for the probability of designation among eligible tracts. The results confirm that governors selected tracts strategically along multiple dimensions. An increase in pre-OZ planning activity increased the probability of designation, as did higher rental shares (relevant for multifamily development), greater employment, and more elastic housing supply. Simultaneously, governors also prioritized designating areas facing economic distress; tracts with lower relative income and higher poverty rates were more likely to be chosen to be opportunity zones.

Columns (2) through (5) present OLS estimates predicting tract-level construction starts after the implementation of the OZ program. The key result is the stark difference between specifications with and without controls for pre-existing planning activity. Without controls (column 2), designated tracts show 0.46 more construction starts than non-

¹¹The average of the state-level variables across tracts are summarized in the bottom rows of Table 1. These variables are similar across tract categories, which is unsurprising given that states were limited to designating 25 percent of eligible tracts as OZs, resulting in fairly equal designation shares across states.

designated eligible tracts. Given that eligible, non-designated tracts only experienced 0.62 construction starts on average, this difference implies roughly 75% more construction starts in designated tracts.

However, this predicted effect falls to 0.18 when we control for pre-OZ planning activity (column 3) and further to 0.15 with the full set of controls (column 4). These results indicate that selection on observables—most prominently, the selection of tracts where more projects were planned in the pre-OZ period—accounts for about two-thirds of the greater construction in OZ designated tracts. The coefficient on pre-OZ planning activity remains large and highly significant across specifications, with each additional planned project before designation associated with 0.21 additional construction starts afterward. This persistence of development momentum independent of OZ designation suggests that much of the construction activity in OZs would have occurred without the program.

Finally, column (5) expands the sample to include all tracts rather than only tracts that were eligible to be OZs. In this analysis, the predicted effect of OZ designation reflects differences in construction starts relative to all non-designated tracts, not just tracts that could have been OZs but were not selected. Coefficient estimates are broadly in line with those in column (4) indicating that patterns identified in the analysis of OZ-eligible tracts extrapolates well to the broader universe of tracts, a fact that will be useful next section when we model outcomes under counterfactual eligibility criteria.

Robustness Table 5 presents estimates from Poisson regressions predicting the number construction starts in a census tract. The model estimates the expected number of construction starts as:

$$\mathbb{E}[\text{Construction Starts}_i] = \exp(\alpha_{s(i)} + \delta \text{OZ Designation}_i + \beta \ln(\text{Pre-OZ Planning}_i) + \gamma X_i),$$

where $\alpha_{s(i)}$ is a state-specific intercept and X_i is the set of tract level controls including the same controls as in Table 2 and an indicator for whether $\text{Pre-OZ Planning}_i = 0$ (in which case we set $\ln(\text{Pre-OZ Planning}_i)$ to 0).¹² This specification allows us to examine proportional changes in construction activity allowing for the presence of 0s.

¹²We replace the pre-OZ Planning variable with the logarithm of pre-OZ Planning since we would expect a linear rather than exponential relationship between plan starts and construction starts. The variable No Pre-OZ planning_i captures differences between tracts with no planned projects and tracts with one.

Without controls (column 1), OZs show 75 percent higher expected construction ($\exp(0.56) - 1$), but this falls to 27 percent with planning controls (column 2) and 22 percent with full controls (column 3)—a proportional reduction similar to our linear specifications.

Table A.2 extends the analysis to alternative outcome measures. For new planning starts (columns 1-3), the OZ effect falls from 0.83 to 0.25 with full controls. For construction completions (columns 4-6), the reduction is from 0.41 to 0.11. The consistency across different stages of the development process strengthens our conclusion that selection on pre-existing development activity accounts for the majority of observed differences between designated and non-designated tracts.

3.2. *Heterogeneity by State*

This section demonstrates that states varied considerably in how they balanced the competing objectives of targeting distressed areas versus areas with development potential, and these differences influence the predicted effect of OZ designation on construction starts. We first validate that OZ designation decisions are aligned with the proxies discussed in Section 2.2. Namely, states with proportional allocation or a low stated prioritization of investment opportunities were less likely to designate tracts with greater planning activity as OZs. We then show that states that prioritized investment potential observed a greater increase in investment in opportunity zones.

Differences in how states selected tracts Table 6 tests whether our state-level measures correlate with states' actual OZ designation decisions. Each specification repeats the analysis from the probit in Table 2, but interacts Pre-OZ Planning_{*i*} with a state-level measure pertaining to how a particular state allocated OZs.¹³

Column (1) shows that proportional selection states were about 30 percent less responsive to Pre-OZ planning when designating tracts (a coefficient of 0.007 compared to 0.010 for non-proportional selection states). However, the difference is statistically insignificant.

Column (2) instead interacts Pre-OZ planning_{*i*} with Investment Opportunity Prioritization_{*i*}, the 1–5 index measuring the extent to which states indicated that investment opportunity was an important consideration in designating tracts. The findings show that states with a

¹³We omit the State Planning Sensitivity measure from this analysis because it mechanically relates to states' tendencies to designate areas with high planning activity.

higher stated prioritization of investment opportunity indeed disproportionately selected tracts with greater development potential as opportunity zones. The point estimates imply designation decisions were roughly unrelated to pre-OZ planning activity for a state at the lowest value for the index (Investment Opportunity Prioritization_{*i*} = 1), while states with a high Investment Opportunity Prioritization_{*i*} = 1 were highly responsive to planning activity. Column (3) shows that relationships are similar when both interaction terms are included together.

Columns (4) and (5) replicate specifications (2) and (3), but replace Investment Opportunity Prioritization_{*i*} = 1 with an indicator for whether it is above 2 (labeled as High Investment Opportunity Prioritization_{*i*} = 1). Findings are similar with this discrete measure; states with a low investment opportunity prioritization generally did not favor tracts with more pre-OZ planning. The relationship between planning and designation decisions is almost entirely driven by states that indicated that they accounted for development potential in selecting opportunity zones.

Altogether, states without proportional allocation rules and states with high investment prioritization scores disproportionately selected tracts with greater pre-OZ planning activity. This set of findings provide validation that our measures effectively capture real differences in state selection strategies that manifested in actual designation decisions.

Differences in the predicted effects of OZ designation Table 7 presents results replicating the specification shown in column (4) of Table 2, but interacting OZ designation with the three state-level selection criteria.

Column (1) shows that states with proportional allocation rules—which limited discretion to concentrate designations in high-growth areas—saw OZ effects roughly half as large as states without such constraints (0.1 vs. 0.21 more construction starts). This difference is consistent with the flexibility to strategically select zones being crucial for generating sizable construction effects, though the question remains whether the results reflect stronger construction effects due to the selection of more responsive markets (heterogeneous effects) or a stronger response due to the selection of tracts that would have had more construction anyway (selection bias).

Column (2) shows similar results for State Planning Sensitivity. A one standard deviation

increase in this measure of a state's tendency to select high planning activity tracts (.03) raises the predicted construction response by .03 projects, a roughly 20 percent larger effect than for states that do not prioritize high planning activity tracts at all. The specification controls for Pre-OZ Planning, so the results do not reflect higher planning activity in designated tracts. Instead, the results suggest that states that selected tracts with greater planning activity also prioritized tracts with greater development potential on unobserved margins. We will account for such selection on unobserved characteristics in the model in the next section.

The most striking results appear in column (3), which uses a GPT-5-generated measure of investment prioritization. States at the median score of 2 see construction increase by only 0.08 projects per designated tract, while each one-point increase in investment prioritization yields an additional 0.11 projects. Tracts in states with the highest investment prioritization score (of 5) are predicted to have 0.47 more construction starts than non-designated tracts—triple the average predicted effect found in Table 2. Meanwhile, states with the lowest score are predicted to have slightly less construction in designated tracts.

Column (4) presents a horse race specification that includes the three interaction terms jointly. The results confirm that both proportional selection rules and low investment prioritization independently dampen the predicted effects of OZ designation. The state-specific planning sensitivity measure becomes irrelevant when these more direct measures are accounted for.

Table 8 provides easier interpretation by using binary measures for planning sensitivity and investment prioritization (i.e., high and low value states). Results are qualitatively similar. States above the median planning sensitivity see nearly twice as much of a construction response in designated tracts (0.20 starts vs. 0.11 starts). Differences are even more stark with investment prioritization: states with scores above 2 see 0.29 more construction starts in designated tracts, while those at or below 2 see essentially no difference (0.04 projects). This seven-fold difference in predicted effects across state groups underscores the pivotal role that selection played in program outcomes.

3.3. *Discussion*

To summarize the reduced form findings, we show that areas designated as opportunity zones saw much more commercial construction than other eligible but not-designated tracts, but a large portion of this difference reflects the selection of tracts with greater development potential as OZs. States that appeared to prioritize development potential more observed larger predicted construction responses to OZ designation.

While these results highlight that selection is important, several questions remain. First, what about selection on unobservables? Though controlling for pre-OZ planning reduces the estimated effects of OZ designation, we still find about 22 percent more construction in OZs controlling for this variable. The true treatment effect of OZ designation is likely smaller to the extent that states designate areas with investment potential on unobserved margins as OZs. Second, why do states see heterogeneous responses to designation? The larger predicted construction response in states prioritizing the selection of markets with development potential could reflect either stronger selection bias or the strategic selection of tracts with a stronger genuine response to designation. Third, what are the tradeoffs in trying to target higher poverty areas? Designating a moderately stressed tract might be more desirable than a severely distressed tract if the latter would have little construction activity regardless of OZ status. Evaluating this tradeoff requires knowledge of how construction might decline if OZs were redirected to higher-poverty tracts. In the next section we present a model of OZ selection and construction activity to address these questions.

4. STRUCTURAL FRAMEWORK

Our reduced-form estimates identify the role of pre-existing development activity in driving measured effects of opportunity zone designation. In this section, we estimate a structural model of OZ designation and subsequent construction activity to provide further context to these estimates and perform relevant counterfactuals.

4.1. Theoretical framework

OZ designation decisions Consider the problem of a state designating opportunity zones. States care about addressing location-specific stresses, and thus want to designate areas with high poverty, but they also want to designate areas that will respond to the program with additional construction. States select opportunity zones from a fixed, state-specific pool of eligible tracts. Their maximization is thus

$$\max_{\{D_i\}} \sum_{i \in I_s} D_i V_i \quad \text{s.t.} \quad \sum_{i \in I_s} D_i \leq Q_s, \quad (1)$$

where D_i is an indicator for whether tract i is designated, I_s is the set of eligible tracts in state s and Q_s is the number of tracts states can select (roughly $0.25 \times |I_s|$, though we take the observed number of tracts selected if it differs). V_i is the utility from designating tract i , which takes the form:

$$V_i = (1 - \theta_{s(i)})\rho \cdot \text{Poverty}_i + \theta_{s(i)}\kappa \cdot \text{DevPotential}_i + \gamma \text{Contig}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, 1). \quad (2)$$

The parameter ρ reflects the baseline preference for targeting poverty (measured as Poverty_i), and κ the baseline preference for development potential (to be defined next). θ_s the relative weight state s places on development opportunity versus poverty, the analog of Investment Opportunity Prioritization in the empirical work. We also include a control for contiguous tracts as a reduced form way of accounting for additional restrictions to designating such tracts.

This framework allows us to give a random utility interpretation to our earlier reduced-form probit (Table 2), where the coefficient estimates on planning activity reflect preferences for development potential and those on poverty reflect preferences for addressing distress.

We next endogenize development potential using the Poisson framework from Table 5; this approach effectively microfound the heterogeneous treatment effects of opportunity zone designation that underlie the preference for areas with higher planning activity.

Endogenizing development potential Suppose that construction start counts in tract i , denoted Y_i , follows a negative binomial process, with an expected number of construction starts:

$$\mathbb{E}[Y_i \mid X_i, D_i, e_i] \equiv \mu(X_i, D_i, e_i) = \exp(\beta_{\text{plan}}^\top X_i + \delta D_i + e_i), \quad (3)$$

where X_i is a vector of variables related to development potential (e.g., previous planning activity), $e_i \sim \mathcal{N}(0, \sigma_y^2)$ is an unobserved development factor that accounts for states potentially designating tracts with unobserved development potential, and δ is the treatment effect of opportunity zone designation. We use a negative binomial process rather than Poisson process to account for overdispersion: $\text{Var}(Y_i) = \mu_i(1 + \alpha\mu_i)$, and we find $\alpha > 0$ whereas it would be 0 for a Poisson process. Accounting for this additional dispersion in construction starts reduces the risk of it being attributed to dispersion in unobserved development potential and causing us to overstate the importance of selection on unobservables.¹⁴

We define development potential as the construction that would be expected to occur without OZ designation:

$$\text{DevPotential}_i = \exp(\beta_{\text{plan}}^\top X_i + e_i). \quad (4)$$

Note that the true treatment effect of OZ designation is $(e^\delta - 1)\text{DevPotential}$, meaning that the effects of designation are proportional to the baseline expected construction in a tract.

We use this structure to estimate the treatment effect of designating an opportunity zone, accounting for selection on both observed (X_i) and unobserved margins (e_i). We additionally use the model to evaluate counterfactual effects of the opportunity zone program under different preferences for development potential relative to economic distress.

¹⁴Empirical results are nearly identical with Poisson and negative binomial regressions.

4.2. Estimation Strategy and Results

We estimate the parameters of the model to maximize the joint likelihood of observing the number of construction starts (Y_i) and designation decisions (D_i) in the data.

Note that the probability of designating a tract as an opportunity zone is:

$$\pi_i(e_i) = \begin{cases} \Phi\left((1 - \theta_{s(i)})\rho\text{Poverty}_i + \theta_{s(i)}\kappa\text{DevPotential}_i + \gamma\text{Contig}_i - v_{s(i)}\right), & \text{if eligible} \\ 0, & \text{if ineligible} \end{cases}$$

where $v_{s(i)}$ is the reservation utility for designating a tract as an OZ in state s . In equilibrium, this is the utility corresponding with the Q_s^{th} highest tract. In the estimation procedure we treat this as a parameter to be estimated (as a state-specific intercept in the probit) but in counterfactuals we account reservation utility being endogenous and adjust it to maintain the same number of designated tracts.

The likelihood of getting the observed construction activity comes from the pmf of a negative binomial distribution (parameterized by its mean and overdispersion), which we denote $f_{\text{NB2}}(Y_i \mid \mu(X_i, D_i, e_i), \alpha)$.

Combining these two expressions provides the likelihood of getting the data for a given unobserved development potential:

$$\mathcal{L}_i(e_i) = \left(f_{\text{NB2}}(Y_i \mid \mu(X_i, D_i = 1, e_i), \alpha)\pi_i(e_i)\right)^{D_i} \left(f_{\text{NB2}}(Y_i \mid \mu(X_i, D_i = 0, e_i), \alpha)(1 - \pi_i(e_i))\right)^{(1-D_i)}$$

Integrating over e_i then gives the likelihood for an individual tract: $\mathcal{L}_i \equiv \int \mathcal{L}_i(e)\phi(e; 0, \sigma_y^2)de$. We approximate the integral over e by Gauss–Hermite quadrature.

One problem with maximizing this directly is that the general level of θ is not well-identified since ρ and κ already reflect prioritization placed on poverty and development potential across states. We effectively normalize θ by assuming a prior over the distribution of θ_s . Specifically, we assume that θ_s comes from a truncated normal distribution that is

centered at 0.5:

$$\theta_s \sim \text{Truncated Normal}(0.5, \sigma_\theta^2; [0, 1]).$$

where σ_θ pertains to the standard deviation of the underlying (not truncated) normal distribution. Imposing this prior accomplishes three things: First, by penalizing deviations from 0.5, this prior provides the needed normalization for the level of θ_s . Second, state-specific sensitivity measures can be noisy, especially in smaller states with fewer eligible tracts, and this approach shrinks noisier estimates towards 0.5 and may prevent incorrect inference of extreme values (Bayesian shrinkage). Third, the approach parameterizes the distribution of θ_s which allows us to consider the effects of resampling preferences rather than taking state-specific estimates for θ as given.

We estimate parameters to maximize the posterior log-likelihood function:

$$\max_{\beta_0, \beta, \delta, \alpha, \rho, \kappa, \gamma, \{\alpha_s\}, \{\theta_s\}, \sigma_y, \sigma_\theta} \sum_i \log \mathcal{L}_i + \sum_s \log p_{\text{TN}}(\theta_s; 0.5, \sigma_\theta, [0, 1]).$$

where p_{TN} is the pdf of a truncated normal distribution.

Parameter estimates Table 9 provides the estimated parameters. The main parameter of interest is δ , which estimates the causal effect of being designated as an opportunity zone on construction. The value of 0.12 means that the effect of being designated as an opportunity zone is a roughly 13 percent increase in expected construction starts. By comparison, the reduced form Poisson regressions estimated about 22 percent expected higher construction. The difference in these estimates is due to the e_i term in the model. Tracts with high unobserved construction potential were more likely to be designated as opportunity zones and would have had higher construction than otherwise similar tracts even if they were not selected. In the estimated model, this bias accounts for a bit less than half of the estimated effect of OZ designation in the reduced form analysis.

The estimated treatment effect is highly sensitive to the dispersion in unobserved development potential (σ_y). A lower σ_y , means that unobserved factors are comparatively less important and estimates for δ are closer to that in the reduced form analysis. This relationship is plotted in Figure 1. Each point corresponds to the estimated effect of OZ designation when we impose a particular value of σ_y ($\sigma_y = 0.7$ corresponds to situation where σ_y is estimated in Table 9). When $\sigma_y = 0$, there is no selection on unobservables,

and the estimate for δ reproduces what we found in the reduced form Poisson regression. As σ_y rises, the unobserved component of development potential becomes more important and the estimated δ falls (i.e., it takes a smaller treatment effect to rationalize the difference in construction outcomes for designated and non-designated tracts). $\hat{\delta}$ falls to around 0 for a σ_y slightly above 1.

Other parameters are roughly in line with reduced form findings. The parameters related to construction starts are similar to the estimates from Poisson regressions. The state preference parameter estimates imply that for a value of $\theta_s = 0.5$, a state would require a development potential that was 0.5 expected starts higher to compensate for a poverty rate that was 1 percentage point lower. This tradeoff reflects a greater preference targeting areas with development potential than was found in the reduced form probit, in part reflecting attenuation bias in the reduced form estimates (since Pre-OZ Planning is just a proxy for development potential).

4.3. Counterfactuals

We now use the estimated model to investigate several counterfactuals that are useful for investigating the open questions discussed in Section 3.3. The main results are in Table 10. The four columns show the average poverty rate for OZ designated tracts, the average Pre-OZ Planning count in OZ designated tracts, the average effect of OZ designation (construction starts caused by the program) in OZ designated tracts, and the raw difference in average construction starts in OZ and eligible, non-OZ tracts. The top line presents averages from the data (excluding treatment effects, which are unobserved) while the rest present results from the model for various OZ selection processes, averaging results over 100 simulations.¹⁵

¹⁵Each simulation makes independent draws for every tract's unobserved development potential (e_i), idiosyncratic utility from being selected (ε_i), and draw from a negative binomial distribution with expectation $\mu(X_i, D_i, e_i)$ where $D_i = 1$ if ε_i is such that V_i is in the $Q_{s(i)}$ highest for the state. In some simulations we resample θ_s in addition to D_i , in which case we additionally pull new state investment preferences from a truncated normal distribution.

Baseline The first three rows show that the model is able to reproduce key patterns in the data. The data generating process in the model produces an average 31% poverty rate in OZ designated tracts (identical to the data), and average Pre-OZ Plan count of 3.82 (a touch below the 3.94 in the data), and a difference in construction between designated and non-designated areas of 0.42 (compared to 0.35 in the data). Averages are also reasonably close to the data if we resample $\{\theta_s\}$ using the estimated distribution of θ across states instead of the state-specific estimates. With redrawn state preferences, we get an average poverty rate of 0.33 and Pre-OZ Plan count of 3.62, indicating that state-specific draws contributed to slightly more prioritization of investment opportunity.

Counterfactual preferences The middle three rows present results for counterfactual state preferences. Each row presents model outcomes if all states have the same investment prioritization of $\theta = 0, 0.5$ or 1.0 .

These counterfactuals achieve two main goals. First, they quantify the cost in terms of construction activity of shifting OZ tax incentives to more highly-distressed regions. The counterfactuals imply that if states ignored investment potential in designating tracts ($\theta = 0$), they would have designated tracts with a 38% average poverty rate. If they fully-prioritized investment potential ($\theta = 1$), the average poverty rate in designated tracts would have only been 26%. This shift towards high poverty markets comes at a cost though; the average effect of OZ designation falls from 0.14 projects for $\theta = 1$, to 0.09 for $\theta = 0$.

Figure 2 plots the relationship between poverty rate and investment response as θ changes more generally, thus tracing out the policy frontier. The figure shows that increasing the average poverty rate by about 1 percentage point results in a construction response that is about 0.005 projects lower.

Second, the counterfactuals shed light on the greater response to OZ designation in high investment prioritization states (since θ_s is the theoretical analogue of the Investment Opportunity Prioritization_s(i) in the empirical work). Conceptually, this differential in response could be attributed to either greater selection bias or a higher treatment on treated. The model allows us to decompose this difference.

The last column shows the raw differences in construction between eligible and designated

tracts (the sum of selection and treatment effects), while the second-to-last isolates the treatment effect. The findings indicate that most of the difference in outcomes for high and low θ reflects selection rather than heterogeneous treatment effects. While going from the lowest to highest prioritization raises the raw difference in construction by 0.55 projects, the treatment effect only goes up by about 0.05. If the data generating process in the model aligns well with that of the data, these results indicate that much of the stark difference in the predicted response of construction to OZ designation reflects greater selection bias in high investment prioritization states. The results also point to potential negative selection bias in areas most focused on targeting poverty: the treatment effect is greater than the raw difference in construction (0.09 vs. 0.05) when $\theta = 0$. Consequently, the low predicted response of construction to designation in low investment prioritization states found in Section 3.2 should not necessarily be taken as indicating that OZs were ineffective in those states. Instead states may have designated tracts that would have had less construction independent of the OZ program.

Alternative eligibility rules The final counterfactual we consider is changing tract eligibility criteria to have more stringent income and poverty thresholds. If national policymakers are more focused on poverty relief than those who designate which eligible tracts are selected, these counterfactuals reflect the ability of policy makers to shift activity to more stressed tracts by changing the menu of eligible tracts. For this counterfactual, we take the revised OZ eligibility criteria in the OBBBA (which moved the income threshold from 80% to 70% of the area median, excluded high poverty tracts if income exceeded 125%, and removed eligibility for contiguous tracts). We then re-simulate outcomes keeping preferences and construction processes the same, but narrowing the set of tracts governors can choose from.

Directionally, the effects of more stringent eligibility criteria are similar to the effects of reducing θ . Relative to the baseline model outcomes, the average poverty rate in designated tracts rises one or two percentage points and the average treatment effect declines by .01 or .02 projects. However, the decline in treatment effects for a given increase in poverty targeting is larger than from reducing θ . Intuitively, a change in preferences to focus more on poverty targeting still allows states to designate some moderately distressed tracts if the opportunity is high enough, while a change in eligibility criteria would remove such states from consideration entirely.

5. CONCLUSION

This paper leverages unique data on pre-existing commercial construction planning activity to show that approximately two-thirds of the observed construction boost in opportunity zones reflects selection of areas with pre-existing development momentum rather than a genuine program impact.

Specifically, our analysis highlights a difficult choice that states faced between targeting truly distressed areas and achieving measurable program results. States that prioritized economic need over development potential saw a smaller response of construction to designation, likely reflecting a combination of reduced selection bias and smaller treatment effects.

We estimate a model of how construction activity is shaped by the opportunity zone program, accounting for the endogenous selection of tracts as OZs and the presence of unobserved tract-level development potential and state-level preferences for OZ designation. We use the model to disentangle selection effects from treatment effects, and investigate several counterfactuals pertaining to program priorities or eligibility criteria.

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Table 1: Summary Statistics

	Not Eligible (1)	Designated (2)	Not Designated (3)	Designated - Not designated (4)
pre-OZ Plan Count	3.45 (6.28)	3.97 (6.69)	2.58 (4.95)	1.39** (0.08)
post-OZ Construction Starts	0.77 (1.94)	1.07 (2.17)	0.61 (1.58)	0.46** (0.03)
post-OZ Plan Count	1.67 (3.83)	2.11 (3.86)	1.26 (3.12)	0.85** (0.05)
post-OZ Construction Completions	0.75 (1.85)	1.00 (2.05)	0.59 (1.53)	0.41** (0.03)
Renter Share	0.26 (0.17)	0.56 (0.22)	0.49 (0.22)	0.07** (0.00)
ln(Employment)	6.71 (1.24)	6.94 (1.30)	6.42 (1.30)	0.53** (0.02)
Housing Supply Elasticity (BH 2024)	0.19 (0.21)	0.10 (0.17)	0.10 (0.18)	-0.01* (0.00)
Missing Supply Elasticity	0.20 (0.40)	0.28 (0.45)	0.23 (0.42)	0.05** (0.01)
Prior Abandonment	0.21 (0.41)	0.23 (0.42)	0.17 (0.37)	0.07** (0.01)
Relative Income	1.21 (0.37)	0.58 (0.21)	0.65 (0.20)	-0.07** (0.00)
Poverty	0.09 (0.05)	0.31 (0.13)	0.26 (0.12)	0.06** (0.00)
ln(Tract Population)	8.32 (0.53)	8.17 (0.54)	8.19 (0.52)	-0.03** (0.01)
Proportional Selection State	0.47 (0.50)	0.47 (0.50)	0.48 (0.50)	-0.00 (0.01)
State Planning Sensitivity	0.02 (0.04)	0.01 (0.04)	0.01 (0.03)	-0.00 (0.00)
State Investment Opportunity Prioritization	2.64 (1.00)	2.53 (0.98)	2.54 (0.98)	-0.01 (0.01)
Democratic Governor	0.42 (0.49)	0.40 (0.49)	0.40 (0.49)	-0.01 (0.01)
Observations	42,055	7,584	23,252	30,836

Note: Columns (1), (2), and (3) present the mean and standard deviation (in parentheses) of various tract characteristics for tracts that were not eligible for opportunity zone designation, tracts that were eligible and designated, and tracts that were eligible but not designated, respectively. Column (4) presents the difference in means, and standard error of the difference, between columns (2) and (3), reflecting differences in tracts that were chosen to be designated vs. those that were not. Relative income is the ratio of the census tract median income to the higher of the relevant MSA or state median income according to the 2015 5-year ACS. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODES workplace area characteristics, and [Baum-Snow and Han \(2024\)](#).

Table 2: Relationship between OZ Designation, Planning Activity, and Construction Starts

	Designated Probit	Construction Starts OLS			
	(1)	(2)	(3)	(4)	(5)
OZ Designation		0.47** (0.08)	0.18** (0.04)	0.15** (0.04)	0.14** (0.04)
Pre-OZ Planning	0.01** (0.00)		0.21** (0.01)	0.21** (0.02)	0.20** (0.01)
Renter Share	0.13 (0.11)			0.27* (0.13)	0.38** (0.12)
ln(Employment)	0.22** (0.03)			0.03 (0.03)	0.04* (0.02)
Housing Supply Elasticity (BH 2024)	0.37+ (0.21)			0.23* (0.09)	0.43** (0.13)
Missing Supply Elasticity	0.43** (0.05)			-0.13** (0.03)	-0.10** (0.03)
Prior Abandonment	0.06** (0.02)			-0.13* (0.06)	-0.14** (0.03)
Relative Income	-0.70** (0.14)			-0.10 (0.09)	-0.05 (0.03)
Poverty	2.08** (0.35)			0.03 (0.14)	-0.10 (0.15)
ln(Tract Population)	-0.13** (0.04)			0.00 (0.03)	0.06* (0.03)
R_a^2		0.055	0.458	0.458	0.453
Observations	30,485	30,836	30,836	30,485	71,617
State FE	✓	✓	✓	✓	✓

Note: Column (1) presents probit estimates of the relationship between OZ designation and tract characteristics, and columns (2)–(5) present OLS estimates predicting tract construction starts based on whether the tract was designated as an OZ, the number of projects planned before the OZ program started, and other tract characteristics. In columns (1)–(4) the sample is tracts that were eligible to be opportunity zones, and column (5) includes the full sample of tracts (thus comparing OZ designated tracts to all other tracts, rather than tracts that were eligible but not chosen). Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODS workplace area characteristics, and [Baum-Snow and Han \(2024\)](#).

Table 3: Relationship between OZ Designation, Planning Activity, and Construction Starts

	Designated Probit	Construction Starts OLS			
	(1)	(2)	(3)	(4)	(5)
OZ Designation		0.15** (0.03)	0.03* (0.01)	0.03* (0.01)	0.03+ (0.02)
Pre-OZ Planning	0.01** (0.00)		0.09** (0.01)	0.09** (0.01)	0.09** (0.01)
Renter Share	0.13 (0.11)			0.17** (0.05)	0.22** (0.06)
ln(Employment)	0.22** (0.03)			-0.00 (0.01)	0.00 (0.01)
Housing Supply Elasticity (BH 2024)	0.37+ (0.21)			0.03 (0.03)	0.07+ (0.04)
Missing Supply Elasticity	0.43** (0.05)			-0.02 (0.01)	-0.01 (0.01)
Prior Abandonment	0.06** (0.02)			-0.08** (0.02)	-0.09** (0.01)
Relative Income	-0.70** (0.14)			0.04 (0.04)	0.01 (0.01)
Poverty	2.08** (0.35)			-0.05 (0.08)	-0.12* (0.06)
ln(Tract Population)	-0.13** (0.04)			-0.01 (0.01)	-0.01 (0.01)
R_a^2		0.043	0.386	0.385	0.377
Observations	30,485	30,836	30,836	30,485	71,617
State FE	✓	✓	✓	✓	✓

Note: Column (1) presents probit estimates of the relationship between OZ designation and tract characteristics, and columns (2)–(5) present OLS estimates predicting tract construction starts based on whether the tract was designated as an OZ, the number of projects planned before the OZ program started, and other tract characteristics. In columns (1)–(4) the sample is tracts that were eligible to be opportunity zones, and column (5) includes the full sample of tracts (thus comparing OZ designated tracts to all other tracts, rather than tracts that were eligible but not chosen). Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODS workplace area characteristics, and [Baum-Snow and Han \(2024\)](#).

Table 4: Relationship between OZ Designation, Planning Activity, and Construction Starts

	Designated Probit	Construction Starts OLS			
	(1)	(2)	(3)	(4)	(5)
OZ Designation		0.46** (0.09)	0.18** (0.04)	0.15** (0.04)	0.15** (0.04)
Pre-OZ Planning	0.01** (0.00)		0.21** (0.01)	0.21** (0.01)	0.20** (0.01)
Renter Share	0.16 (0.12)			0.34** (0.13)	0.38** (0.12)
ln(Employment)	0.21** (0.03)			0.03 (0.02)	0.04* (0.02)
Housing Supply Elasticity (BH 2024)	0.40* (0.19)			0.25* (0.11)	0.43** (0.13)
Missing Supply Elasticity	0.41** (0.05)			-0.10** (0.03)	-0.10** (0.03)
Prior Abandonment	0.06** (0.02)			-0.14** (0.05)	-0.14** (0.03)
Relative Income	-0.66** (0.13)			-0.05 (0.09)	-0.05 (0.04)
Poverty	2.08** (0.35)			0.01 (0.15)	-0.09 (0.16)
Contiguous Tract	-0.96** (0.10)			0.06** (0.02)	0.02 (0.02)
ln(Tract Population)	-0.14** (0.04)			0.01 (0.03)	0.06* (0.03)
R_a^2		0.047	0.458	0.460	0.453
Observations	40,728	41,085	41,085	40,728	71,617
State FE	✓	✓	✓	✓	✓

Note: Column (1) presents probit estimates of the relationship between OZ designation and tract characteristics, and columns (2)–(5) present OLS estimates predicting tract construction starts based on whether the tract was designated as an OZ, the number of projects planned before the OZ program started, and other tract characteristics. In columns (1)–(4) the sample is tracts that were eligible to be opportunity zones, and column (5) includes the full sample of tracts (thus comparing OZ designated tracts to all other tracts, rather than tracts that were eligible but not chosen). Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODES workplace area characteristics, and [Baum-Snow and Han \(2024\)](#).

Table 5: Poisson Estimates for Construction Starts

	Construction Starts			
	(1)	(2)	(3)	(4)
OZ Designation	0.57** (0.09)	0.24** (0.04)	0.21** (0.04)	0.20** (0.03)
ln(Pre-OZ Planning)		0.89** (0.02)	0.79** (0.04)	0.79** (0.04)
No Pre-OZ Planning		-0.84** (0.08)	-0.72** (0.07)	-0.72** (0.05)
Renter Share			0.49** (0.15)	0.51** (0.12)
ln(Employment)			0.10* (0.05)	0.13** (0.04)
Housing Supply Elasticity (BH 2024)			0.39** (0.10)	0.63** (0.12)
Missing Supply Elasticity			-0.45** (0.06)	-0.32** (0.05)
Prior Abandonment			-0.05 (0.04)	-0.04 (0.04)
Relative Income			-0.04 (0.05)	-0.06 (0.04)
Poverty			0.08 (0.14)	-0.17 (0.15)
ln(Tract Population)			0.06** (0.02)	0.13** (0.01)
R_p^2	0.078	0.363	0.371	0.375
Observations	30,836	30,836	30,485	71,617
State FE	✓	✓	✓	✓

Note: Each column presents Poisson estimates of the relationship between construction starts and tract characteristics. Columns (1)–(4) included the same variables as Columns (2)–(5) in Table 2 but with a Poisson regression rather than OLS. The only change is that the pre-OZ plan starts is replaced with two variables: the logarithm of pre-OZ plan starts and an indicator for whether there were no pre-OZ plan starts (in which case we set the logarithm to 0). Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODES workplace area characteristics, Frank et al. (2022), and Baum-Snow and Han (2024).

Table 6: OZ Designation by State Selection Characteristics

	Designated				
	(1)	(2)	(3)	(4)	(5)
Pre-OZ Planning	0.010** (0.002)	-0.009* (0.004)	-0.008+ (0.004)	0.002 (0.002)	0.003 (0.003)
...×Proportional Selection	-0.002 (0.003)		-0.002 (0.003)		-0.001 (0.003)
...×Investment Opportunity Prioritization		0.007** (0.001)	0.007** (0.001)		
...×High Investment Opportunity Prioritization				0.012** (0.003)	0.012** (0.003)
R_a^2					
Observations	30,485	30,485	30,485	30,485	30,485
Controls	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓

Note: This table repeats the probit from specification (1) from Table 2, but interacts Pre-OZ planning with state characteristics pertaining to how tracts within a state were designated. Column (1)'s state measure (Proportional Selection) is the interaction with whether states used a proportional allocation to designate tracts, using data from Frank et al. (2022). Column (2)'s state measure (Investment Opportunity Prioritization) is a score (from 1 to 5) indexing how much a state took into account investment potential in designating tracts (as formed by GPT-5's scoring of public comments regarding allocation decisions). Column (3) presents results with both measures included simultaneously. Columns (4) and (5) repeat the specifications of (2) and (3), respectively, but using an indicator for whether the Investment Opportunity Prioritization score is above 2 in place of the continuous measure. The same controls as in Table 2 are included in all specifications, but not displayed. Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODS workplace area characteristics, Frank et al. (2022), and Baum-Snow and Han (2024).

Table 7: Differential Construction Response by State Selection Characteristics

	Construction Starts			
	(1)	(2)	(3)	(4)
OZ Designation	0.20** (0.02)	0.15** (0.02)	-0.11* (0.05)	-0.07 (0.05)
...×Proportional Selection	-0.11** (0.03)			-0.09* (0.03)
...×State Planning Sensitivity		0.59 (0.49)		-0.40 (0.52)
...×State Investment Opportunity Prioritization			0.11** (0.02)	0.11** (0.02)
Pre-OZ Planning	0.21** (0.00)	0.21** (0.00)	0.21** (0.00)	0.21** (0.00)
Renter Share	0.27** (0.05)	0.27** (0.05)	0.27** (0.05)	0.27** (0.05)
ln(Employment)	0.03** (0.01)	0.03** (0.01)	0.03** (0.01)	0.03** (0.01)
Housing Supply Elasticity (BH 2024)	0.23** (0.05)	0.23** (0.05)	0.24** (0.05)	0.24** (0.05)
Missing Supply Elasticity	-0.14** (0.02)	-0.13** (0.02)	-0.13** (0.02)	-0.13** (0.02)
Prior Abandonment	-0.13** (0.02)	-0.13** (0.02)	-0.13** (0.02)	-0.13** (0.02)
Relative Income	-0.10* (0.05)	-0.10* (0.05)	-0.10* (0.05)	-0.10* (0.05)
Poverty	0.04 (0.08)	0.04 (0.08)	0.04 (0.08)	0.04 (0.08)
ln(Tract Population)	0.01 (0.02)	0.00 (0.02)	0.01 (0.02)	0.01 (0.02)
R_a^2	0.459	0.458	0.459	0.459
Observations	30,485	30,485	30,485	30,485
State FE	✓	✓	✓	✓

Note: This table repeats specification (4) from Table 2, but interacts the OZ Designation indicator with state characteristics pertaining to how tracts within a state were designated. Column (1) adds an interaction with whether states used a proportional allocation to designate tracts, using data from [Frank et al. \(2022\)](#). Column (2) adds an interaction with the imputed weight that a state placed on initial planning activity in selecting tracts, namely the state specific coefficient estimate on Pre-OZ Planning when the specification from Column (1) of Table 2 is expanded to interaction of Pre-OZ Planning with state indicator variables. Column (3) adds the a score indexing how much a state took into account investment potential in designating tracts (as formed by GPT-5's scoring of public comments regarding allocation decisions). Column (4) presents results with all three interactions included simultaneously. Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODS workplace area characteristics, [Frank et al. \(2022\)](#), and [Baum-Snow and Han \(2024\)](#).

Table 8: Differential Construction Response by State Selection Characteristics, Discrete Measures

	Construction Starts			
	(1)	(2)	(3)	(4)
OZ Designation	0.20** (0.02)	0.12** (0.02)	0.05* (0.02)	0.11** (0.03)
...×Proportional Selection	-0.11** (0.03)			-0.07* (0.04)
...×High Planning Sensitivity		0.07+ (0.03)		-0.05 (0.04)
...×High Investment Opportunity Prioritization			0.23** (0.03)	0.23** (0.04)
Pre-OZ Planning	0.21** (0.00)	0.21** (0.00)	0.21** (0.00)	0.21** (0.00)
Renter Share	0.27** (0.05)	0.27** (0.05)	0.27** (0.05)	0.27** (0.05)
ln(Employment)	0.03** (0.01)	0.03** (0.01)	0.03** (0.01)	0.03** (0.01)
Housing Supply Elasticity (BH 2024)	0.23** (0.05)	0.23** (0.05)	0.24** (0.05)	0.24** (0.05)
Missing Supply Elasticity	-0.14** (0.02)	-0.13** (0.02)	-0.13** (0.02)	-0.13** (0.02)
Prior Abandonment	-0.13** (0.02)	-0.13** (0.02)	-0.13** (0.02)	-0.13** (0.02)
Relative Income	-0.10* (0.05)	-0.10* (0.05)	-0.10* (0.05)	-0.10* (0.05)
Poverty	0.04 (0.08)	0.04 (0.08)	0.04 (0.08)	0.04 (0.08)
ln(Tract Population)	0.01 (0.02)	0.01 (0.02)	0.01 (0.02)	0.01 (0.02)
R_a^2	0.459	0.459	0.459	0.459
Observations	30,485	30,485	30,485	30,485
State FE	✓	✓	✓	✓

Note: This table repeats specification (4) from Table 2, but interacts the OZ Designation indicator with an indicator for whether state characteristics pertaining to how tracts within a state were designated (as described in Section 2 and shown in Table A.1) above (High) their median value. Column (1)'s state measure (Proportional Selection) is whether states used a proportional allocation to designate tracts, using data from Frank et al. (2022). Column (2)'s state measure (Investment Opportunity Prioritization) is the imputed weight that a state placed on initial planning activity in selecting tracts, namely the state specific coefficient estimate on Pre-OZ Planning when the specification from Column (1) of Table 2 is expanded to include the interaction of Pre-OZ Planning with state indicator variables. Column (3)'s state measure (Investment Opportunity Prioritization) is a score (from 1 to 5) indexing how much a state took into account investment potential in designating tracts (as formed by GPT-5's scoring of public comments regarding allocation decisions); for this measure, the value is denoted as High if its value is above 2. Column (4) presents results with all three measures included simultaneously. Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODS workplace area characteristics, Frank et al. (2022), and Baum-Snow and Han (2024).

Table 9: Estimated Parameters

Parameter	Description	Estimate
Construction parameters		
δ	Proportional Effect of Designation	0.12
β_0	Intercept	-1.59
β_1	Coefficient on ln(Pre-OZ Planning)	0.92
β_2	Coefficient on No Pre-OZ Planning	-0.88
Selection parameters		
ρ	Poverty Preference	8.03
κ	Development Preference	0.16
γ	Contiguous tract preference	-1.09
Distributional parameters		
σ_y	SD of unobserved development factor	0.70
α_{NB}	NB overdispersion	0.19
σ_θ	SD of state prioritization θ_s (TN prior)	1.70

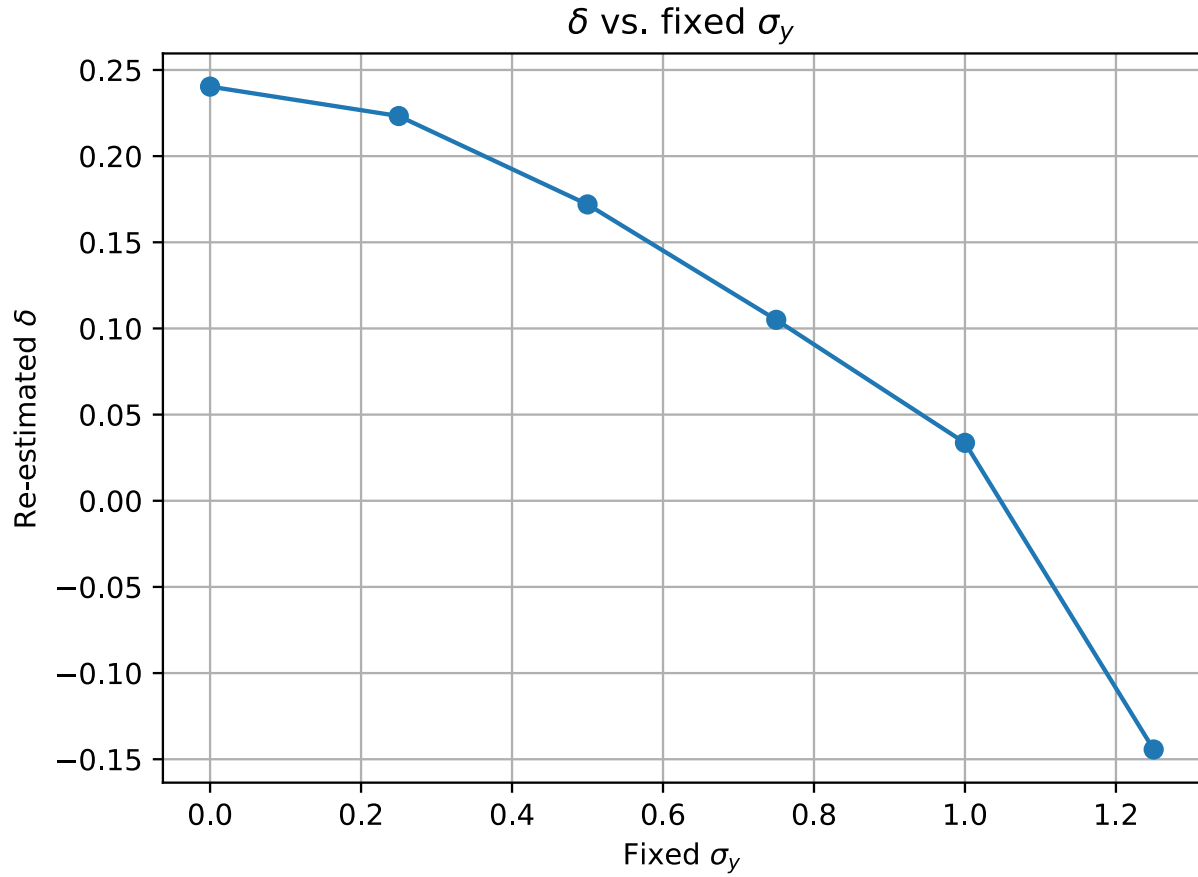
Note: The table shows parameter estimates from the model presented in Section 4.

Table 10: Counterfactual Outcomes

Scenario	Avg. Poverty	Avg. Plan Stock	Treatment on Treated	Raw Diff. Construction
Observed data	0.31	3.94		0.35
Model DGP (estimated params)	0.31	3.82	0.13	0.42
Model DGP (redrawn θ_s)	0.33	3.62	0.12	0.32
Counterfactual Investment Prioritization (θ)				
$\theta_s \equiv 0$	0.38	3.08	0.09	0.05
$\theta_s \equiv 0.5$	0.33	3.60	0.12	0.31
$\theta_s \equiv 1$	0.26	4.22	0.14	0.60
Counterfactual Eligibility Criteria				
Alt. eligibility rule	0.33	3.57	0.11	0.30
Alt. eligibility (redrawn θ_s)	0.34	3.41	0.11	0.23

Note: Each row gives the average poverty rate in designated tracts, the average Pre-OZ Plan Count in designated tracts, the causal effect of OZ designation (in construction start counts), and the raw difference in construction starts between designated and non-designated tracts for a given scenario. The top set of rows provide averages in the observed data (where available), the averages from the data generating process implied by the model at the estimated values of $\{\theta_s\}$, and the model when θ_s are redrawn from the estimated distribution. The next block presents counterfactuals when governors have uniform preferences with $\theta = 0$, $\theta = .5$, and $\theta = 1$. The last block presents counterfactuals for the revised income and poverty eligibility criteria in the OBBBA. *Source:* Authors' calculations using Dodge Data and Analytics and estimates from the simulated model described in Section 4.

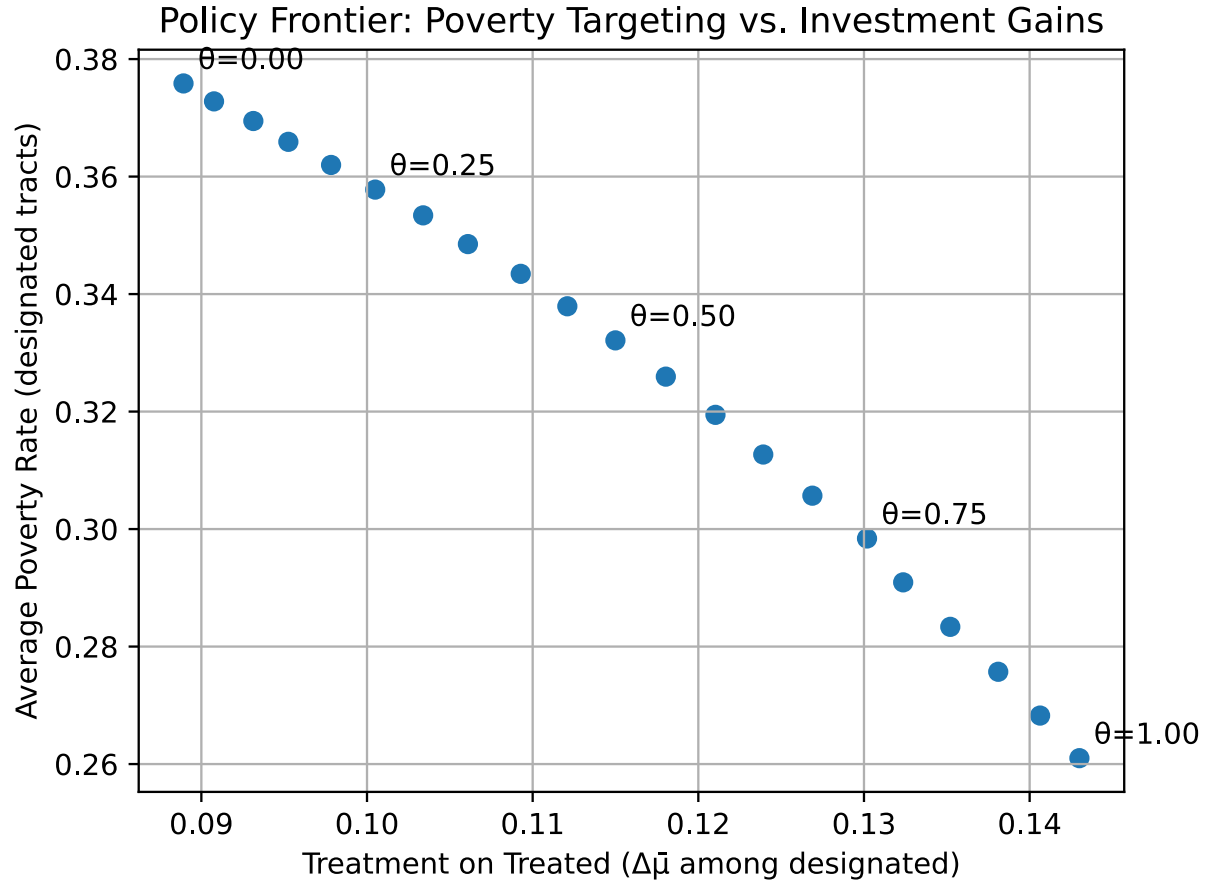
Figure 1: Sensitivity of $\hat{\delta}$ to σ_y



Note: This is a plot of estimates of the proportional effect of OZ designation ($\hat{\delta}$) conditional on the volatility of unobserved development potential (σ_y). Each point gives the $\hat{\delta}$ estimate when σ_y is set to the value on the x-axis rather than estimated. Besides fixing σ_y , the estimation procedure is the same as in Section 4.

Source: Authors' calculations.

Figure 2: Aid vs. Investment Tradeoffs



Note: This is a plot of the tradeoff between designating tracts with high poverty vs. designating tracts that will respond more to OZ designation. Each point gives the counterfactual average poverty rate in designated areas (y-axis) and treatment effect of OZ designation in designated areas (x-axis) when states have different uniform prioritization of investment (θ). Dots move down and to the right (lower poverty and higher investment response) as θ rises. *Source:* Authors' calculations.

A. ADDITIONAL TABLES AND FIGURES REFERENCED IN THE MAIN TEXT

Table A.1: Measures of State OZ Selection Criteria

State	Proportional Selection	Planning Sensitivity	Investment Opportunity Prioritization
Alabama	1	.041	1
Alaska	0	-.02	2
Arizona	1	.036	3
Arkansas	0	.008	3
California	1	-.004	2
Colorado	0	.007	3
Connecticut	0	.06	3
Delaware	0	.051	2
Florida	1	-.013	3
Georgia	0	-.036	2
Hawaii	0	.186	2
Idaho	0	-.012	2
Illinois	1	-.047	2
Indiana	0	.021	5
Iowa	0	-.003	2
Kansas	1	.049	2
Kentucky	1	0	2
Louisiana	0	.031	2
Maine	0	.027	3
Maryland	1	.037	2
Massachusetts	0	.009	5
Michigan	1	.039	2
Minnesota	0	.002	2
Mississippi	0	.053	2
Missouri	1	.052	3
Montana	0	-.121	2
Nebraska	0	.13	2
Nevada	0	.06	5
New Hampshire	0	.014	2
New Jersey	1	.073	4
New Mexico	0	.057	3
New York	0	.024	3
North Carolina	1	.005	2
North Dakota	0	-.006	2
Ohio	0	.029	3
Oklahoma	0	.04	2
Oregon	0	.083	2
Pennsylvania	0	.017	4
Rhode Island	0	.072	2
South Carolina	1	-.006	2
South Dakota	0	.031	2
Tennessee	0	.004	3
Texas	0	-.033	1
Utah	0	.02	3
Vermont	0	.173	3
Virginia	1	.037	3
Washington	1	.005	5
West Virginia	0	.083	2
Wisconsin	0	.051	3
Wyoming	0	.094	2

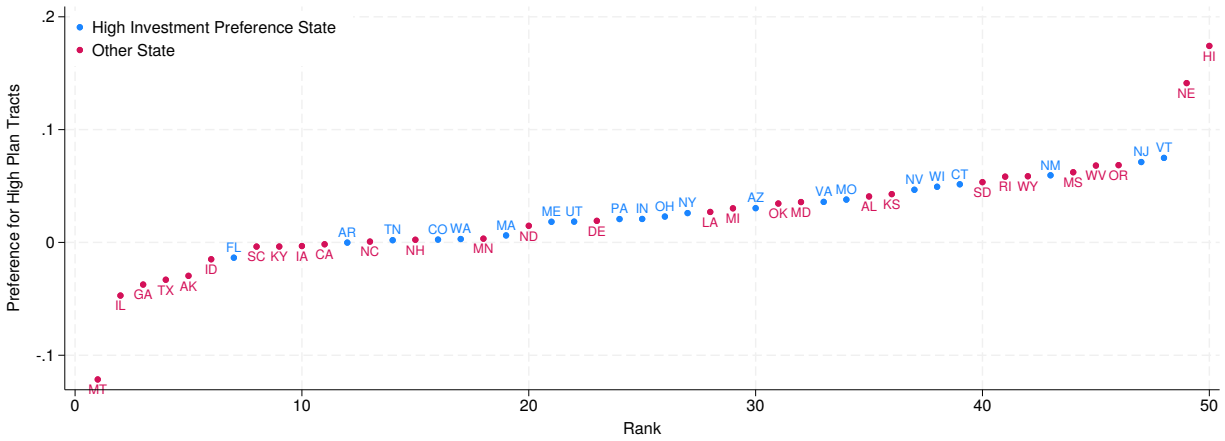
Note: This table presents measures of state OZ selection criteria. Proportional Selection is whether states used a proportional allocation to designate tracts, using data from [Frank et al. \(2022\)](#). Planning Sensitivity is the imputed weight that a state placed on initial planning activity in selecting tracts, namely the state specific coefficient estimate on Pre-OZ Planning when the specification from Column (1) of Table 2 is expanded to include the interaction of Pre-OZ Planning with state indicator variables. Investment Opportunity Prioritization is a score (from 1 to 5) indexing how much a state took into account investment potential in designating tracts (as formed by GPT-5's scoring of public comments regarding allocation decisions). *Source:* Authors' calculations using data from [Frank et al. \(2022\)](#), Dodge Data and Analytics, ACS, LODES workplace area characteristics, [Baum-Snow and Han \(2024\)](#), and GPT-5.

Table A.2: Relationship between OZ Designation, Planning Activity, and Planning Starts or Construction Completions

	Planning Starts			Construction Completions		
	(1)	(2)	(3)	(4)	(5)	(6)
OZ Designation	0.86** (0.13)	0.35** (0.06)	0.26** (0.06)	0.42** (0.08)	0.12** (0.03)	0.11** (0.03)
Pre-OZ Planning		0.36** (0.02)	0.34** (0.02)		0.21** (0.01)	0.21** (0.02)
Renter Share			0.14 (0.14)			0.32* (0.13)
ln(Employment)			0.16** (0.02)			0.01 (0.03)
Housing Supply Elasticity (BH 2024)			0.77** (0.23)			0.17* (0.07)
Missing Supply Elasticity			-0.30** (0.06)			-0.09** (0.03)
Prior Abandonment			-0.00 (0.10)			-0.14** (0.05)
Relative Income			-0.44** (0.16)			-0.05 (0.10)
Poverty			0.08 (0.21)			-0.02 (0.15)
ln(Tract Population)			0.07+ (0.04)			-0.00 (0.03)
R_a^2	0.052	0.388	0.390	0.055	0.497	0.498
Observations	30,836	30,836	30,485	30,836	30,836	30,485
State FE	✓	✓	✓	✓	✓	✓

Note: Columns (1)–(3) replicate Columns (2)–(4) in Table 2 but for a new dependent variable: planning starts. Columns (4)–(6) perform a similar exercise for a new dependent variable: construction completions. Standard errors are clustered at the state level. +, *, and ** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively. *Source:* Authors’ calculations using Dodge Data and Analytics, ACS, LODES workplace area characteristics, and [Baum-Snow and Han \(2024\)](#).

Figure A.1: State Planning Sensitivity



Note: Figure plots the the state-specific coefficient estimate on Pre-OZ Planning when the specification from Column (1) of Table 2 is expanded to the interaction of Pre-OZ Planning with state Planning Sensitivity. Planning Sensitivity is the imputed weight that a state placed on initial planning activity in selecting tracts, namely the state specific coefficient estimate on Pre-OZ Planning when the specification from Column (1) of Table 2 is expanded to include the interaction of Pre-OZ Planning with state indicator variables. Investment Opportunity Prioritization is a score (from 1 to 5) indexing how much a state took into account investment potential in designating tracts (as formed by GPT-5's scoring of public comments regarding allocation decisions). High Investment Opportunity Prioritization states are denoted by blue dots with labels above markers, while other states are denoted by red dots, with labels below the markers. *Source:* Authors' calculations using Dodge Data and Analytics, ACS, LODS workplace area characteristics, and Baum-Snow and Han (2024).